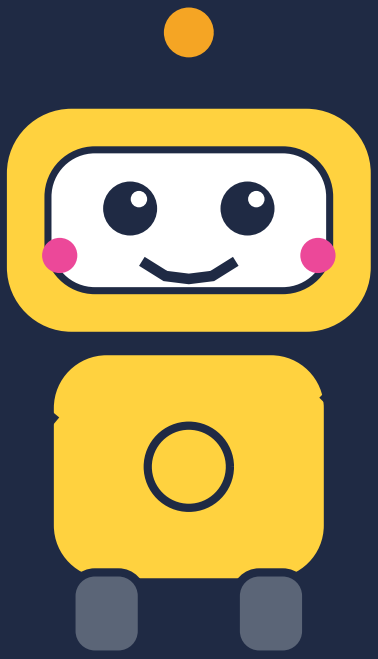
TEACHER BOOK

Think Like a Coder

Teacher book for Class 7

Period-by-period plans for every CT and AI chapter
Teacher guides for both 20-hour projects
Answer keys, assessment guide and photocopiable journal



For CBSE's Computational Thinking and AI curriculum, 2026-27

Goes with Think Like a Coder, Class 7 · Follows NCERT Class 7 maths, CBSE's four AI chapters and two projects

SAMPLE EDITION · DRAFT

© 2026 Bullseye Knowledge System Limited. All rights reserved.
CodeSchool is a working name. Not an official CBSE or NCERT publication.

What's inside

About this teacher book . . . 3
How a lesson runs . . . 4
Year plan . . . 5
Assessment . . . 6
1. Large Numbers Around Us . . . 7
2. Arithmetic Expressions . . . 9
3. A Peek Beyond the Point . . . 11
4. Expressions using Letter-Numbers . . . 13
5. Parallel and Intersecting Lines . . . 15
6. Number Play . . . 17
7. A Tale of Three Intersecting Lines . . . 19
8. Working with Fractions . . . 21
AI 1. AI Domains and Applications . . . 23
AI 2. AI in Industries . . . 26
AI 3. Data Visualisation and Analysis . . . 29
AI 4. Ethics and AI Bias Awareness . . . 32
Project 1. Nutri-Quest Food Planner . . . 35
Project 2. Sort It with AI . . . 39
Observation journal (photocopy) . . . 43
Term summary (photocopy) . . . 44

© 2026 Bullseye Knowledge System Limited

About this teacher book

This book goes with the *Think Like a Coder* Class 7 workbook. It gives you a plan for every chapter, so that a maths or class teacher can teach computational thinking and AI confidently without being a computer specialist. CBSE suggests about **100 hours** a year for Class 7; the plans here use **150 periods of 40 minutes**, which is about 100 hours.

★ What is in this book

- ✓ **A year plan** showing which chapter to teach in which term, next to the NCERT maths chapter.
- ✓ **A period-by-period plan for every chapter**, including the AI chapters: what you do and what students do.
- ✓ **A hands-on activity** for every chapter, with set-up steps and a tip.
- ✓ **Common mistakes** to watch for, and how to support or stretch students.
- ✓ **An exit check** with answers, and what to note in the observation journal.
- ✓ **The answer key** for every page of the student workbook.
- ✓ **Photocopiable** observation journal and term summary pages at the back.
- ✓ **A teacher guide for each 20-hour project**: week-by-week plan, roles, set-up, pitfalls and how to mark with the rubric.

✓ Before you start

- ✓ Read the chapter in the student book and these notes. Most lessons need 5 to 10 minutes to prepare.
- ✓ Get the kit items ready (listed under *At a glance*). Everything is reusable.
- ✓ Plan pairs in advance: mix confident readers with children who prefer talking and acting out.
- ✓ Teach the CT chapter in the same week or fortnight as the matching NCERT maths chapter, so the maths is fresh.
- ✓ Keep it active: children should talk, act out and build more than they listen.

Computers are optional

No computer lab is needed. Every CT and AI idea is taught with paper, cards and talk, and both projects run on paper. Where computers are available, the optional boxes in the AI chapters and projects suggest short demonstrations or tools. A projector helps but is optional.

© 2026 Bullseye Knowledge System Limited

How a lesson runs

Every lesson fits one 40-minute period and follows the same five steps, based on the PRIMM approach used in computing classrooms (predict, run, investigate, modify, make). Children learn the routine once and then focus on the thinking.

1. **Predict and run** 8 min
   Children predict what a set of steps will do, then act it out.
2. **Investigate** 8 min
   Pairs answer 'why' questions about how it worked.
3. **Modify** 8 min
   Children change the steps to do something new.
4. **Put in order** 8 min
   Jumbled code cards go back in the right order.
5. **Make** 8 min
   Pairs write their own program and swap with another pair.

How it supports CBSE's assessment

CBSE asks schools to assess CT&AI through written puzzles, group activities and a **Teacher Observation Journal**. Each step gives the teacher something to observe: predictions show reasoning, modifications show understanding, and the 'make' step shows whether a child can build an algorithm of their own. The journal format and a term summary for the report card come with the teacher book.

Every child can take part

- ✓ Children who finish early move to the challenge level; children who need help use a hint card.
- ✓ Most activities are done in pairs, so children who find reading hard can still take part by talking and acting out.
- ✓ Teacher preparation is about 5-10 minutes per lesson using the script.

© 2026 Bullseye Knowledge System Limited

Year plan

8 CT chapters of 7 periods plus 4 review periods (about 40 hours), 4 AI chapters of 7 periods plus 2 review periods (about 20 hours), and two projects of 30 periods each (about 40 hours): **150 periods** of 40 minutes in all. Teach each chapter alongside the NCERT maths chapter with the same name; the order and timing can follow your school's maths plan.

Ch	Chapter	Coder skill	Level	Periods	Term
1	Large Numbers Around Us	Compare, round and estimate big numbers step by step	C4	7	Term 1
2	Arithmetic Expressions	Order of operations and expression trees	C4	7	Term 1
3	A Peek Beyond the Point	Decimals, binary numbers and punched-card sorting	C4	7	Term 1
4	Expressions using Letter-Numbers	Variables and formulas for growing patterns	C4	7	Term 1
5	Parallel and Intersecting Lines	Angle rules and tracing drawing programs	C5	7	Term 2
6	Number Play	True or false: list every case and filter with clues	C5	7	Term 2
7	A Tale of Three Intersecting Lines	Triangle rules as checks; debugging a function	C5	7	Term 2
8	Working with Fractions	Fraction operations and working backwards	C4	7	Term 2
AI1	AI Domains and Applications	Classify, predict a number, or find groups	C5	7	Term 1
AI2	AI in Industries	AI in health, education, transport and communication	C5	7	Term 1
AI3	Data Visualisation and Analysis	Choose bar, line or pie charts and read them honestly	C5	7	Term 2
AI4	Ethics and AI Bias Awareness	Spot bias in data and keep humans in charge	C5	7	Term 2
P1	Nutri-Quest Food Planner	Use data and rules to plan a balanced week	C5	30	Term 1
P2	Sort It with AI	Build a waste classifier and measure its accuracy	C5	30	Term 2
	Revision and term assessment	Review games, the term summary and parent feedback	All	6	Both

© 2026 Bullseye Knowledge System Limited

Assessment and the observation journal

CBSE asks for continuous, competency-based assessment in CT&AI: written puzzles, group activities and a **Teacher Observation Journal**. This book makes that simple. In each chapter, the exit check and the 'what to note' box tell you what to look for; record one level per child per skill every few chapters, not every lesson.

Skill	Beginning	Developing	Secure	Extending
Algorithms and formulas	Follows given steps	Evaluates expressions in the right order	Writes formulas and traces programs	Designs and debugs checks
Logic	Uses one clue	Filters cases with two clues	Lists every case and filters them	Explains why a rule always holds
AI ideas	Names AI domains	Tells classification, regression, clustering apart	Measures accuracy on test data	Finds and explains bias in data
Projects	Takes part with help	Completes stages with help	Plans, tests and presents with the team	Improves the work using evidence

How to assess without extra tests

- **Watch and listen** during activities: explaining, predicting and fixing show understanding better than a written answer.
- **Use the exit check** at the end of each chapter: two quick questions, answers in this book.
- **Compare with self-checks**: each chapter ends with 'How did I do?', where children colour a face for each 'I can' statement.
- **Summarise each term** on the term summary page: one level per area and a short comment for the report card.

The four levels

Use the four levels for every skill. **Beginning**: needs help to start. **Developing**: succeeds with prompts or a partner. **Secure**: succeeds alone on familiar problems. **Extending**: solves new problems and explains why. Most children will be Developing or Secure; that is expected and healthy.

© 2026 Bullseye Knowledge System Limited

1

TEACHER NOTES · CHAPTER 1 · LEVEL C4

Large Numbers Around Us

Coder skill: Compare, round and estimate big numbers step by step

Goes with NCERT maths, Class 7, chapter 1



At a glance	
Students will	read lakhs, crores and millions, compare big numbers step by step and round and estimate.
Key words	**lakh** (1,00,000), **crore** (1,00,00,000), **million** (10,00,000 (10 lakh)), **estimate** (a close, easy answer)
Materials	Number cards with big numbers (populations, distances, budgets) in Indian and international forms, place-value charts
Time	7 periods of 40 minutes
Link	Teach alongside chapter 1 of the NCERT Class 7 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *lakh*, *crore*, *million*, *estimate*. Work through the example *Which is bigger: 4,56,78,912 or 45,678,921?* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Big number line-up	Run the activity below. Move between groups and note who can explain the idea.	Convert big numbers between systems, order them with the compare algorithm, and round to estimate.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	An algorithm that compares two big numbers. Model the first card or step, then let pairs finish.	Trace it for 9,87,654 and 10,02,003. Then for 45,67,890 and 45,69,001. In which place is the first difference?
5	Practise and extend	Go over the questions students found hardest, then run the stretch task with the whole class.	Correct their answers, explain one fix to a partner, then try the stretch task.
6	Puzzle period	Set two or three puzzles from the matching chapter of CBSE's Class 7 handbook. Pairs solve them, then present one method.	Solve handbook puzzles in pairs and explain one method to the class.
7	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

© 2026 Bullseye Knowledge System Limited

Hands-on activity: Big number line-up

1. Give each group 6 cards with big numbers, some written with Indian commas and some with international commas.
2. Groups rewrite each number in both systems on a place-value chart.
3. They order the cards using the compare algorithm: count digits first, then compare from the left.
4. Groups round each number to the nearest lakh and to the nearest crore, and estimate a total.

Tip: Ask 'Which place decided it?' after each comparison. Naming the place shows the algorithm at work.

Common mistakes to watch for

- **Reading commas wrongly**: 4,56,78,912 is 4 crore 56 lakh, not 456 million.
- **Comparing from the right**, or ignoring the number of digits.
- **Rounding to the wrong place**: the nearest lakh looks at the ten-thousands digit.

Different paces

Support: Use 6- and 7-digit numbers in the Indian system only.
Stretch: How many lakhs make 1 billion? (10,000) Write the steps to convert any crore value to millions.

Exit check and what to note

Two quick questions at the end of the chapter:

1. Write 3,45,67,890 in the international system. (Answer: 34,567,890)

2. Round 7,48,52,000 to the nearest crore. (Answer: 7,00,00,000)

Note in the journal: Beginning (needs help to read lakhs, crores and millions); **Developing** (can read lakhs, crores and millions with prompts or a partner); **Secure** (can read lakhs, crores and millions and compare big numbers step by step alone); **Extending** (can round and estimate and explain why).

Answer key

1. 100
2. b) 10 lakh
3. 7,73,300 and 3,00,377 (a number cannot start with 0)
4. 4,87,00,000
5. c) 0: anything multiplied by 0 is 0
Build it: 10,02,003 has 7 digits against 6, so it is bigger. 45,69,001 is bigger: the first difference is in the thousands place (7 against 9)
Challenge: 1,40,00,00,000 = 1,400,000,000, which is 1.4 billion

© 2026 Bullseye Knowledge System Limited

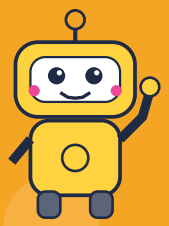
2

TEACHER NOTES · CHAPTER 2 · LEVEL C4

Arithmetic Expressions

Coder skill: Order of operations and expression trees

Goes with NCERT maths, Class 7, chapter 2



At a glance	
Students will	follow the order of operations, use brackets to change the order and read an expression tree.
Key words	**expression** (numbers joined by operations), **term** (a part added or subtracted), **brackets** (do this part first), **precedence** (which operation goes first)
Materials	Number cards and operator cards (+, -, x, ÷, brackets) large enough to hold up, string for tree branches
Time	7 periods of 40 minutes
Link	Teach alongside chapter 2 of the NCERT Class 7 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *expression*, *term*, *brackets*, *precedence*. Work through the example *Work out 8 + 6 x (5 - 2)* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Human expression trees	Run the activity below. Move between groups and note who can explain the idea.	Build and evaluate expression trees, write expressions from trees, and use brackets to hit a target.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	A program that adds up a shopping bill. Model the first card or step, then let pairs finish.	What does it say? Write the whole bill as one expression.
5	Practise and extend	Go over the questions students found hardest, then run the stretch task with the whole class.	Correct their answers, explain one fix to a partner, then try the stretch task.
6	Puzzle period	Set two or three puzzles from the matching chapter of CBSE's Class 7 handbook. Pairs solve them, then present one method.	Solve handbook puzzles in pairs and explain one method to the class.
7	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

© 2026 Bullseye Knowledge System Limited

Hands-on activity: Human expression trees

1. Write 8 + 6 x (5 - 2) on the board. Students hold number and operator cards to form its tree, joined with string.
2. Work from the bottom: the students holding 5, - and 2 'become' 3, and so on up to the top.
3. Groups build trees for their own expressions and swap with another group to evaluate.
4. Play 'make 24' with 2, 3, 4 and 5, using brackets.

Tip: The operator at the top of the tree is done last. Ask 'Which card goes at the top?' before building.

Common mistakes to watch for

- **Working left to right** and ignoring the order of operations.
- **Doing + before ÷** when no brackets say so.
- **Brackets in the wrong place** when writing a tree as an expression.

Different paces

Support: Use expressions with only two operations.
Stretch: Write one expression whose value changes in four different ways depending on where brackets go.

Exit check and what to note

Two quick questions at the end of the chapter:

1. Work out 18 - 6 ÷ 3 x 2. (Answer: 14 (6 ÷ 3 x 2 = 4))

2. Work out (4 + 5) x (8 - 6). (Answer: 18)

Note in the journal: Beginning (needs help to follow the order of operations); **Developing** (can follow the order of operations with prompts or a partner); **Secure** (can follow the order of operations and use brackets to change the order alone); **Extending** (can read an expression tree and explain why).

Answer key

1. 8 and 48
2. c) x (6 x 4 + 10 = 34)
3. 8 (23 - 5 + 8 - 6 + 12 = 32 passengers)
4. a) (2 + 3) x 4
5. 40: (3 + 5) x (9 - 4)
Build it: 3 x 45 + 2 x 120 - 50 = 135 + 240 - 50 = 325
Challenge: For example 4 x (5 + 3 - 2) = 24 or 2 x (3 + 4 + 5) = 24; accept any correct expression

© 2026 Bullseye Knowledge System Limited

3

TEACHER NOTES · CHAPTER 3 · LEVEL C4

A Peek Beyond the Point

Coder skill: Decimals, binary numbers and punched-card sorting

Goes with NCERT maths, Class 7, chapter 3



At a glance	
Students will	read decimal place values, change numbers between binary and decimal and explain how 5 bits can label 26 letters.
Key words	**decimal** (place values in tens), **binary** (place values in twos), **bit** (a binary digit, 0 or 1), **place value** (what a digit's position is worth)
Materials	Binary dot cards (16, 8, 4, 2, 1), punched-card letter deck and sorting sticks (kit), decimal place-value charts
Time	7 periods of 40 minutes
Link	Teach alongside chapter 3 of the NCERT Class 7 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *decimal*, *binary*, *bit*, *place value*. Work through the example *Writing 23 in binary with dot cards* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Binary cards and punched-card sorting	Run the activity below. Move between groups and note who can explain the idea.	Show numbers with binary dot cards, read binary patterns, and sort letter cards with punched holes.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	An algorithm that turns a decimal number into binary by halving. Model the first card or step, then let pairs finish.	Trace it for 14. Write each value of n and each remainder. What binary number do you get?
5	Practise and extend	Go over the questions students found hardest, then run the stretch task with the whole class.	Correct their answers, explain one fix to a partner, then try the stretch task.
6	Puzzle period	Set two or three puzzles from the matching chapter of CBSE's Class 7 handbook. Pairs solve them, then present one method.	Solve handbook puzzles in pairs and explain one method to the class.
7	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

© 2026 Bullseye Knowledge System Limited

Hands-on activity: Binary cards and punched-card sorting

1. Five students hold the dot cards 16, 8, 4, 2, 1. Call a number; they show it by turning cards face up (1) or down (0).
2. Reverse it: show a pattern and the class says the number.
3. Groups shuffle the punched letter cards and sort them with sticks, starting from the rightmost hole, as in CBSE's handbook activity.
4. Ask: 'Why are 5 holes enough for 26 letters?' (5 bits give 32 patterns.)

Tip: Let students discover that the biggest number with 5 cards is 31 before you tell them.

Common mistakes to watch for

- **Reading binary from the wrong end**: the rightmost bit is the 1s place.
- **Missing a zero** in the middle, such as writing 10011 as 1011.
- **Comparing decimals by length**: 0.5 is bigger than 0.405.

Different paces

Support: Use 3 cards (4, 2, 1) for numbers up to 7.
Stretch: Write your age, your birth year and today's date in binary. Which needs the most bits?

Exit check and what to note

Two quick questions at the end of the chapter:

1. What is 11 in binary? (Answer: 1011)

2. What is the binary number 11001 in decimal? (Answer: 25)

Note in the journal: Beginning (needs help to read decimal place values); **Developing** (can read decimal place values with prompts or a partner); **Secure** (can read decimal place values and change numbers between binary and decimal alone); **Extending** (can explain how 5 bits can label 26 letters and explain why).

Answer key

1. b) 1101 (8 + 4 + 1)
2. 18 (16 + 2)
3. b) 6: 5 bits give only 32 numbers, 6 bits give 64
4. c) 0.405
5. a) 11.0 (alternately x 2 and + 0.2)
Build it: 14: remainder 0, n = 7; remainder 1, n = 3; remainder 1, n = 1; remainder 1, n = 0. Read backwards: 1110
Challenge: 10011 = 19 = S; R = 18 = 10010

© 2026 Bullseye Knowledge System Limited

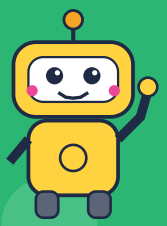
TEACHER NOTES · CHAPTER 4 · LEVEL C4

4

Expressions using Letter-Numbers

Coder skill: Variables and formulas for growing patterns

Goes with NCERT maths, Class 7, chapter 4



At a glance	
Students will	use letters for numbers, substitute values into a formula and write a formula for a growing pattern.
Key words	**variable** (a letter that stands for a number), **formula** (a rule written with letters), **substitute** (put a number in place of a letter), **like terms** (terms with the same letter)
Materials	Matchsticks or straws, squared paper, pattern tables, pyramid puzzle sheets
Time	7 periods of 40 minutes
Link	Teach alongside chapter 4 of the NCERT Class 7 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *variable*, *formula*, *substitute*, *like terms*. Work through the example *Matchstick squares in a row* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Formula factory	Run the activity below. Move between groups and note who can explain the idea.	Build growing patterns, write formulas with variables, and substitute to predict far-off steps.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	A program that uses the matchstick formula. Model the first card or step, then let pairs finish.	Run it for n = 10 and for n = 17. What does it say each time?
5	Practise and extend	Go over the questions students found hardest, then run the stretch task with the whole class.	Correct their answers, explain one fix to a partner, then try the stretch task.
6	Puzzle period	Set two or three puzzles from the matching chapter of CBSE's Class 7 handbook. Pairs solve them, then present one method.	Solve handbook puzzles in pairs and explain one method to the class.
7	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

© 2026 Bullseye Knowledge System Limited

Hands-on activity: Formula factory

1. Groups build matchstick squares in a row (1, 2, 3, 4 squares) and record the sticks.
2. They write the formula 3n + 1 and test it on every row of the table.
3. Groups invent a new growing pattern (triangles, houses or L-shapes) and write its formula.
4. Swap patterns: another group substitutes n = 20 and checks by reasoning, not drawing.

Tip: Ask 'What does the 3 mean, and what does the 1 mean?' Linking each part to the picture builds understanding.

Common mistakes to watch for

- **Writing 3n + 1 as 31n** or confusing 3n with 3 + n.
- **Testing the formula on one step only.**
- **Adding unlike terms**, such as 3x + 2 = 5x.

Different paces

Support: Use patterns that grow by 2 and give the formula's first part.
Stretch: Find the formula for houses in a row (5, 9, 13 sticks) and the number of houses that use exactly 81 sticks (20).

Exit check and what to note

Two quick questions at the end of the chapter:

1. If a = 3 and b = 5, what is 4a - b? (Answer: 7)

2. Squares in a row need 3n + 1 sticks. How many squares can you make with exactly 31 sticks? (Answer: 10)

Note in the journal: Beginning (needs help to use letters for numbers); **Developing** (can use letters for numbers with prompts or a partner); **Secure** (can use letters for numbers and substitute values into a formula alone); **Extending** (can write a formula for a growing pattern and explain why).

Answer key

1. 14 (20 - 6)
2. a) 5x + 4
3. c) 3 (B = 9, Z = 12)
4. 8: the middle row is 3 + P and P + 4, so 2P + 7 = 23
5. a) 14p (2p + 12p)
Build it: n = 10: 31 sticks. n = 17: 52 sticks, and it says 'Buy another box'
Challenge: 2n + 1; 51 sticks

© 2026 Bullseye Knowledge System Limited

5

TEACHER NOTES · CHAPTER 5 · LEVEL C5

Parallel and Intersecting Lines

Coder skill: Angle rules and tracing drawing programs

Goes with NCERT maths, Class 7, chapter 5



At a glance	
Students will	use vertically opposite and straight-line angles, spot parallel lines in a drawing program and find angles made by a transversal.
Key words	**parallel** (never meet), **intersect** (cross at a point), **vertically opposite** (equal angles across a crossing), **transversal** (a line crossing two lines)
Materials	Masking tape for two parallel lines and a transversal on the floor, angle cards, protractors, squared paper
Time	7 periods of 40 minutes
Link	Teach alongside chapter 5 of the NCERT Class 7 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *parallel*, *intersect*, *vertically opposite*, *transversal*. Work through the example *Two lines cross, and one angle is 65°* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Angle hunt	Run the activity below. Move between groups and note who can explain the idea.	Find equal angles at a transversal using rules, and spot parallel lines in a turtle program.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Trace this drawing program on dotted paper. Model the first card or step, then let pairs finish.	What shape does it draw? Then write a program for a rhombus with sides of 60 whose opposite sides are parallel.
5	Practise and extend	Go over the questions students found hardest, then run the stretch task with the whole class.	Correct their answers, explain one fix to a partner, then try the stretch task.
6	Puzzle period	Set two or three puzzles from the matching chapter of CBSE's Class 7 handbook. Pairs solve them, then present one method.	Solve handbook puzzles in pairs and explain one method to the class.
7	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

© 2026 Bullseye Knowledge System Limited

Hands-on activity: Angle hunt

1. Tape two parallel lines and a transversal on the floor, making 8 angles.
2. Measure one angle with a protractor. Students stand in angles they think are equal to it.
3. The class checks each claim with a rule: vertically opposite, corresponding or alternate.
4. On squared paper, pairs trace a turtle program and mark which lines are parallel by following the heading.

Tip: Give each angle a number from 1 to 8 so students can say 'angle 3 equals angle 5 because ...'.

Common mistakes to watch for

- **Measuring when a rule would do**; use rules first, then check.
- **Mixing up corresponding and alternate angles.**
- **Thinking any two lines that look parallel are parallel.** Check the angles or the heading.

Different paces

Support: Use only vertically opposite angles and straight-line angles first.
Stretch: A turtle draws a parallelogram with turns of 70° and 110°. Write the program.

Exit check and what to note

Two quick questions at the end of the chapter:

1. At a transversal across parallel lines, one angle is 70°. What sizes are the other angles? (Answer: 70° and 110° (four of each))

2. Vertically opposite angles are (2x + 10)° and 50°. What is x? (Answer: 20)

Note in the journal: Beginning (needs help to use vertically opposite and straight-line angles); **Developing** (can use vertically opposite and straight-line angles with prompts or a partner); **Secure** (can use vertically opposite and straight-line angles and spot parallel lines in a drawing program alone); **Extending** (can find angles made by a transversal and explain why).

Answer key

1. 108°
2. x = 20; each angle is 70°
3. a) 3:00 and 9:00
4. a) parallel (the heading has turned 180°)
5. c) 30 (16 + 9 + 4 + 1)
Build it: An equilateral triangle. Rhombus: repeat 2 times: forward 60, turn right 60°, forward 60, turn right 120°
Challenge: 4 angles are 50° and the other 4 are 130°

© 2026 Bullseye Knowledge System Limited

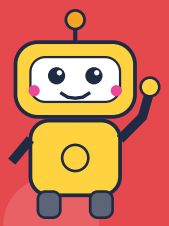
6

TEACHER NOTES · CHAPTER 6 · LEVEL C5

Number Play

Coder skill: True or false: list every case and filter with clues

Goes with NCERT maths, Class 7, chapter 6



At a glance	
Students will	list all true/false combinations, use clues to filter cases and explain why a rule is always true.
Key words	**true or false** (the two values in logic), **case** (one possible combination), **filter** (remove cases that break a rule), **truth table** (a list of every case)
Materials	Punched-card logic deck for R, G and B (8 cards) and sorting sticks (kit), clue cards
Time	7 periods of 40 minutes
Link	Teach alongside chapter 6 of the NCERT Class 7 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *true or false*, *case*, *filter*, *truth table*. Work through the example *Three lights R, G and B: which are ON?* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Punched-card logic	Run the activity below. Move between groups and note who can explain the idea.	Filter a deck of all ON/OFF combinations clue by clue, and write puzzles with one solution.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	A computer can solve a number puzzle by checking every case. Model the first card or step, then let pairs finish.	Which numbers does the program say? Find them without checking all 90 numbers.
5	Practise and extend	Go over the questions students found hardest, then run the stretch task with the whole class.	Correct their answers, explain one fix to a partner, then try the stretch task.
6	Puzzle period	Set two or three puzzles from the matching chapter of CBSE's Class 7 handbook. Pairs solve them, then present one method.	Solve handbook puzzles in pairs and explain one method to the class.
7	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

© 2026 Bullseye Knowledge System Limited

Hands-on activity: Punched-card logic

1. Each group gets the 8 RGB cards. Check that every ON/OFF combination is there once.
2. Read clue 1. Groups use the stick in the right hole to lift out (or keep) cards that break the clue.
3. Continue clue by clue until one card is left. Record how many cards were left after each clue.
4. Groups write their own three-clue puzzle with one solution and swap.

Tip: This is CBSE's handbook activity. Practise the stick method once yourself so you know which cards lift out.

Common mistakes to watch for

- **Removing the wrong cards**: for 'Red is ON', remove the cards where Red is OFF.
- **Losing track of earlier clues.** Every remaining card must fit all clues so far.
- **Puzzles with no solution or many solutions.** Test them before swapping.

Different paces

Support: Use 2 lights (4 cards) first.
Stretch: Use the 32-card deck for 5 lights and design a puzzle that needs exactly 4 clues.

Exit check and what to note

Two quick questions at the end of the chapter:

1. How many cards are in the deck for 5 lights? (Answer: 32)

2. Clues: Red is OFF, Blue is ON, exactly one light is ON. Which card is left? (Answer: OOU: only Blue is on)

Note in the journal: Beginning (needs help to list all true/false combinations); **Developing** (can list all true/false combinations with prompts or a partner); **Secure** (can list all true/false combinations and use clues to filter cases alone); **Extending** (can explain why a rule is always true and explain why).

Answer key

1. 16
2. b) 4
3. a) all ON (UUU)
4. c) 9, because (10a + b) - (10b + a) = 9(a - b)
5. b) no: two odd digits make an even sum, and even digits keep it even
Build it: Only 74: 9 x (a - b) = 27 means a - b = 3, and a + b = 11, so a = 7 and b = 4 (74 - 47 = 27)
Challenge: 32 cards; 3 are left (Y on, P off, and two of R, G and B on)

© 2026 Bullseye Knowledge System Limited

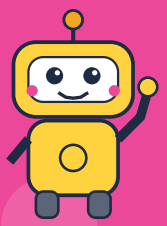
7

TEACHER NOTES · CHAPTER 7 · LEVEL C5

A Tale of Three Intersecting Lines

Coder skill: Triangle rules as checks; debugging a function

Goes with NCERT maths, Class 7, chapter 7



At a glance	
Students will	test the triangle inequality, use the angle sum of 180° and debug a checking function.
Key words	**triangle inequality** (each side < sum of the other two), **equilateral** (all sides equal), **isosceles** (two sides equal), **scalene** (no sides equal)
Materials	Drinking straws cut to 3, 4, 5, 6, 8 and 10 cm, rulers, protractors, results tables
Time	7 periods of 40 minutes
Link	Teach alongside chapter 7 of the NCERT Class 7 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *triangle inequality*, *equilateral*, *isosceles*, *scalene*. Work through the example *Can these lengths make a triangle?* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Straw triangles	Run the activity below. Move between groups and note who can explain the idea.	Test straw triples, discover the triangle inequality, and fix a checking function that tests only one condition.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	This function is meant to check whether three lengths make a triangle. Model the first card or step, then let pairs finish.	Test it with 2, 10, 3. What does it give back, and what is the right answer? Fix the bug.
5	Practise and extend	Go over the questions students found hardest, then run the stretch task with the whole class.	Correct their answers, explain one fix to a partner, then try the stretch task.
6	Puzzle period	Set two or three puzzles from the matching chapter of CBSE's Class 7 handbook. Pairs solve them, then present one method.	Solve handbook puzzles in pairs and explain one method to the class.
7	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

© 2026 Bullseye Knowledge System Limited

Hands-on activity: Straw triangles

1. Groups pick three straws at a time and try to make a triangle, recording 'yes' or 'no' in a table.
2. After 8 tries, groups look for a rule that predicts 'yes'. Guide them to the triangle inequality.
3. Check the rule on two new triples before trying them.
4. Groups test the buggy is_triangle(a, b, c) card from the book with their 'no' triples and fix it.

Tip: Include a triple where the two short sides exactly equal the long side (3, 5, 8). It looks close but lies flat.

Common mistakes to watch for

- **Checking only one pair of sides.** The shortcut is: the two shortest must add up to more than the longest.
- **Counting equal sums as a triangle**: 3 + 5 = 8 does not make one.
- **Angle sum slips**: the three angles add up to 180°, not 360°.

Different paces

Support: Give triples already sorted from shortest to longest.
Stretch: List all triangles with whole-number sides and a perimeter of 15 cm, and classify each. (7: 1-7-7, 2-6-7, 3-5-7, 4-4-7, 3-6-6, 4-5-6, 5-5-5)

Exit check and what to note

Two quick questions at the end of the chapter:

1. Can 4 cm, 6 cm and 11 cm make a triangle? (Answer: no: 4 + 6 = 10 is less than 11)

2. Two angles of a triangle are 35° and 85°. What is the third? (Answer: 60°)

Note in the journal: Beginning (needs help to test the triangle inequality); **Developing** (can test the triangle inequality with prompts or a partner); **Secure** (can test the triangle inequality and use the angle sum of 180° alone); **Extending** (can debug a checking function and explain why).

Answer key

1. c) 5, 7, 10
2. 65°
3. b) right-angled
4. b) 2: 6, 6, 8 and 8, 8, 4 (4, 4, 12 and 2, 2, 16 break the triangle inequality)
5. a) the square (16 cm^2, the triangle is about 6.9 cm^2)
Build it: It says 'yes' (2 + 10 > 3), but 2 + 3 < 10, so it is not a triangle. Fix: If a + b > c AND b + c > a AND a + c > b
Challenge: 3: (2, 5, 5), (3, 4, 5) and (4, 4, 4)

© 2026 Bullseye Knowledge System Limited

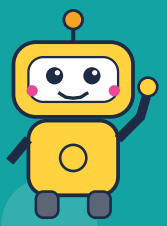
8

TEACHER NOTES · CHAPTER 8 · LEVEL C4

Working with Fractions

Coder skill: Fraction operations and working backwards

Goes with NCERT maths, Class 7, chapter 8



At a glance	
Students will	find a fraction of a quantity, multiply and divide fractions and work backwards to find a start value.
Key words	**of** (means multiply), **reciprocal** (the fraction turned upside down), **inverse** (the step that undoes another), **work backwards** (start from the end)
Materials	Fraction program cards, counters, small whiteboards
Time	7 periods of 40 minutes
Link	Teach alongside chapter 8 of the NCERT Class 7 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *of*, *reciprocal*, *inverse*, *work backwards*. Work through the example *Meera's sweets* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: The undo machine	Run the activity below. Move between groups and note who can explain the idea.	Run fraction programs, then work backwards with inverse steps to recover the start number.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	A fraction function used inside a program. Model the first card or step, then let pairs finish.	Trace it. What is money after each step, and what does the program say?
5	Practise and extend	Go over the questions students found hardest, then run the stretch task with the whole class.	Correct their answers, explain one fix to a partner, then try the stretch task.
6	Puzzle period	Set two or three puzzles from the matching chapter of CBSE's Class 7 handbook. Pairs solve them, then present one method.	Solve handbook puzzles in pairs and explain one method to the class.
7	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

© 2026 Bullseye Knowledge System Limited

Hands-on activity: The undo machine

1. Student A secretly chooses a start number and runs a card program on it, such as *take away 1/3, then take away 1/4 of what is left.*
2. A says only the final answer.
3. Student B works backwards with inverse steps, last step first, to find the start number.
4. Swap roles. Ask: 'Why do we undo the last step first?'

Tip: Choose start numbers divisible by 12 at first so every step gives a whole number.

Common mistakes to watch for

- **Undoing in the original order** instead of last step first.
- **Using the wrong inverse**: 'take away 1/3' is undone by dividing by 2/3, not adding 1/3 of the answer.
- **Dividing by a fraction** without turning it upside down.

Different paces

Support: Use one-step programs such as 'take away half'.
Stretch: Design a three-step fraction program whose undo needs a division by 3/5.

Exit check and what to note

Two quick questions at the end of the chapter:

1. What is 2/5 of 45? (Answer: 18)

2. After spending 1/4 of her money, Asha has Rs 120 left. How much did she have at the start? (Answer: Rs 160)

Note in the journal: Beginning (needs help to find a fraction of a quantity); **Developing** (can find a fraction of a quantity with prompts or a partner); **Secure** (can find a fraction of a quantity and multiply and divide fractions alone); **Extending** (can work backwards to find a start value and explain why).

Answer key

1. 36
2. a) 3/5 (18/30)
3. 80: before B there were 80; before the 20 got on, 60; that is 3/4 of the start
4. a) 8
5. 1/5
Build it: 600, then 600 - 200 = 400, then 400 - 100 = 300: it says 300
Challenge: Set money to 300; divide money by 3/4 (400); divide money by 2/3 (600). Undo the last step first

© 2026 Bullseye Knowledge System Limited

AI1

TEACHER NOTES · AI CHAPTER 1 · LEVEL C5

AI Domains and Applications

AI skill: Classify, predict a number, or find groups

Goes with the AI part of CBSE's Class 7 CT&AI handbook, chapter 1



At a glance	
Students will	name three AI domains, tell classification, regression and clustering apart and explain training, validation and test data.
Key words	**classification** (sort into labelled groups), **regression** (predict a number), **clustering** (find groups without labels), **dataset** (the data a model learns from)
Materials	Scenario cards (spam filter, house prices, photo grouping, crop yield), fruit cards with size and sweetness, a floor grid for a scatter plot
Time	7 periods of 40 minutes
Link	Goes with the AI part of CBSE's Class 7 CT&AI handbook, chapter 1.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *classification*, *regression*, *clustering*, *dataset*. Work through the example *Predicting a number from a small dataset (regression)* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Classify, predict or cluster?	Run the activity below. Move between groups and note who can explain the idea.	Sort AI examples by technique and domain, and act out clustering on a floor scatter plot.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Be a clustering algorithm. Each dot is a fruit, placed by its size and sweetness. There are no labels. Model the first card or step, then let pairs finish.	Circle the groups you find. How many are there, and what could each group be?
5	Prediction line	Plot hours of practice against marks on the board. Students draw a line by eye and predict a new value, then discuss how sure they are.	Predict a number from a small dataset and explain why it is only an estimate.
6	Train, validate, test	Split a deck of 20 cards into 12 training, 4 validation and 4 test cards. Ask why the test cards must stay unseen.	Split a dataset and explain the job of each part.
7	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

© 2026 Bullseye Knowledge System Limited

Hands-on activity: Classify, predict or cluster?

1. Groups sort scenario cards into classification, regression and clustering, and into data science, computer vision and NLP.
2. Each group explains one card using the question: does it give a label, a number or a group?
3. Make a floor scatter plot: students place fruit cards by size (across) and sweetness (up).
4. The class circles the clusters it sees, without any labels, then names each group.

Tip: Keep the domain and the technique separate: a camera app can classify (vision) and a sales app can predict numbers (data).

Common mistakes to watch for

- **Calling every AI task 'classification'.** Ask: is the answer a label, a number or a group?
- **Mixing up domains and techniques.**
- **Using test data during training**, which makes the score look better than it is.

Different paces

Support: Sort only 3 clear scenario cards, one per technique.
Stretch: Give one example of each technique from a single app, such as a music app.

If you have computers (optional)

Use a spreadsheet chart with a trend line to predict marks from practice hours, or a no-code tool to train a two-class image model.

Exit check and what to note

Two quick questions at the end of the chapter:

1. Predicting house prices from size: which technique? (Answer: regression)

2. Grouping songs by how they sound, with no labels: which technique? (Answer: clustering)

Note in the journal: Beginning (needs help to name three AI domains); **Developing** (can name three AI domains with prompts or a partner); **Secure** (can name three AI domains and tell classification, regression and clustering apart alone); **Extending** (can explain training, validation and test data and explain why).

© 2026 Bullseye Knowledge System Limited

Answer key

1. b) regression
2. c) clustering
3. b) Computer Vision
4. 75
5. c) NLP
Build it: Two groups: small, very sweet fruit (like grapes or berries) and big, less sweet fruit (like apples or guavas). Clustering finds them without labels
Challenge: Open answer, for example: sorting library books by subject (classification); predicting lunch numbers (regression); grouping students by favourite clubs (clustering)

© 2026 Bullseye Knowledge System Limited

AI2

TEACHER NOTES · AI CHAPTER 2 · LEVEL C5

AI in Industries

AI skill: AI in health, education, transport and communication

Goes with the AI part of CBSE's Class 7 CT&AI handbook, chapter 2



At a glance	
Students will	give examples of AI in four industries, explain how a route app decides and say why humans stay responsible.
Key words	**diagnosis** (finding out what illness it is), **personalised** (made for one person), **route planning** (choosing the best way), **translation** (changing one language to another)
Materials	Sector cards (healthcare, education, transport, communication), case-study sheet, traffic-light rule cards
Time	7 periods of 40 minutes
Link	Goes with the AI part of CBSE's Class 7 CT&AI handbook, chapter 2.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *diagnosis*, *personalised*, *route planning*, *translation*. Work through the example *How a route app chooses* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: AI job fair	Run the activity below. Move between groups and note who can explain the idea.	Present how AI helps one sector with a benefit and a risk, and act out rule order in a smart traffic light.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Rules for a smart traffic light at a crossing of two roads. Model the first card or step, then let pairs finish.	Road 1 has 25 cars and road 2 has 8. What happens? Then an ambulance arrives on road 2. What happens now?
5	Case study	Read a short case, such as AI translating distress calls in several languages for a rescue team. Groups list the data the AI needed.	Identify the data behind an AI system in a real case.
6	Who is responsible?	Hold a short debate: 'If a route app sends a driver into flooding, who is responsible?'	Argue a position with reasons and listen to the other side.
7	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

© 2026 Bullseye Knowledge System Limited

Hands-on activity: AI job fair

1. Each group takes a sector card and prepares a 2-minute 'stall': one way AI helps, one benefit and one risk.
2. Groups visit each other's stalls and note one question.
3. Act out the smart traffic light: students are cars on two roads; a 'controller' applies the car-count and ambulance rules.
4. Ask: 'Which rule must come first? What happens if it does not?'

Tip: Insist on a risk for every benefit. Responsible use is part of the CBSE learning outcome.

Common mistakes to watch for

- **Only listing benefits.** Ask for a risk or limit each time.
- **Thinking AI replaces doctors or teachers.** It supports their decisions.
- **Putting the ambulance rule last** in the traffic program.

Different paces

Support: Give each group a sentence frame: 'AI helps ___ by ___. A risk is ___.'
Stretch: Design rules for a smart school bell that adjusts for exam days and assemblies.

If you have computers (optional)

Try a translation app on a sentence in a regional language and back again. Discuss what changed and why.

Exit check and what to note

Two quick questions at the end of the chapter:

1. AI that reads X-rays helps which sector? (Answer: healthcare)

2. Why should a doctor check an AI diagnosis? (Answer: AI can make mistakes, and the doctor is responsible for the patient)

Note in the journal: Beginning (needs help to give examples of AI in four industries); **Developing** (can give examples of AI in four industries with prompts or a partner); **Secure** (can give examples of AI in four industries and explain how a route app decides alone); **Extending** (can say why humans stay responsible and explain why).

© 2026 Bullseye Knowledge System Limited

Answer key

1. a) healthcare
2. b) education
3. c) transport
4. c) C (10 km), even though it is the slowest
5. a) AI can make mistakes, and the doctor is responsible for the patient
Build it: Road 1 gets a longer green light; when the ambulance is detected, road 2 turns green at once
Challenge: The ambulance check, because an emergency must win over any queue; checked last, the car rule could delay it

© 2026 Bullseye Knowledge System Limited

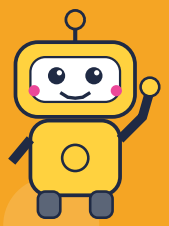
AI3

TEACHER NOTES · AI CHAPTER 3 · LEVEL C5

Data Visualisation and Analysis

AI skill: Choose bar, line or pie charts and read them honestly

Goes with the AI part of CBSE's Class 7 CT&AI handbook, chapter 3



At a glance	
Students will	choose the right chart, work out pie chart angles and read a trend and check that a chart is honest.
Key words	**structured data** (data in rows and columns), **line graph** (shows change over time), **pie chart** (shows parts of a whole), **accuracy** (closeness to the true value)
Materials	Survey sheets, squared paper, protractors, a week of temperature data, two kitchen scales (optional)
Time	7 periods of 40 minutes
Link	Goes with the AI part of CBSE's Class 7 CT&AI handbook, chapter 3.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *structured data*, *line graph*, *pie chart*, *accuracy*. Work through the example *How 40 students travel to school* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: One dataset, three charts	Run the activity below. Move between groups and note who can explain the idea.	Turn class data into bar, pie and line charts and decide which chart fits which question.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	A shop's ice-cream sales from Monday to Sunday: 12, 15, 14, 20, 26, 40, 38. Model the first card or step, then let pairs finish.	Draw the chart in the space and write the trend.
5	Honest or misleading?	Show the same data with an axis starting at 0 and one starting at 90. Students explain which is honest.	Spot and fix a misleading chart.
6	Precision and accuracy	Weigh the same object on two scales (or use sample readings). Discuss precise, accurate, both or neither.	Classify measurements as precise, accurate, both or neither.
7	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

© 2026 Bullseye Knowledge System Limited

Hands-on activity: One dataset, three charts

1. The class collects data: how students travel to school (for a bar and a pie chart).
2. Groups draw a bar chart, then work out pie-chart angles (count ÷ total x 360°) and draw the pie chart.
3. Give a week of temperatures; groups draw a line graph.
4. Ask: 'Which chart answers which question best?'

Tip: Check that pie angles add up to 360° before drawing. It catches most arithmetic slips.

Common mistakes to watch for

- **Pie angles that do not add up to 360°.**
- **Line graphs for categories** that are not in order, such as favourite fruits.
- **Axes without labels or units.**

Different paces

Support: Give pie data that divides 360 easily, such as a total of 36 or 60.
Stretch: Find a chart in a newspaper or textbook and judge whether it is honest.

If you have computers (optional)

Make all three charts in a spreadsheet from the class data, and compare them with the hand-drawn ones.

Exit check and what to note

Two quick questions at the end of the chapter:

1. In a pie chart, 30 of 120 students chose cricket. What angle is the cricket slice? (Answer: 90°)

2. Which chart is best for showing change over time? (Answer: a line graph)

Note in the journal: Beginning (needs help to choose the right chart); **Developing** (can choose the right chart with prompts or a partner); **Secure** (can choose the right chart and work out pie chart angles alone); **Extending** (can read a trend and check that a chart is honest and explain why).

© 2026 Bullseye Knowledge System Limited

Answer key

1. b) line graph
2. a) pie chart
3. 90° (15/60 x 360)
4. a) precise but not accurate
5. a) a table of marks
Build it: A line graph. Sales rise through the week and are highest at the weekend (40 on Saturday)
Challenge: The bar heights no longer show the real values; start the axis at 0, or clearly mark the break in the axis

© 2026 Bullseye Knowledge System Limited

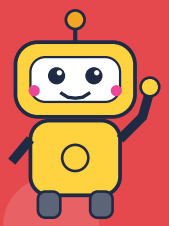
AI4

TEACHER NOTES · AI CHAPTER 4 · LEVEL C5

Ethics and AI Bias Awareness

AI skill: Spot bias in data and keep humans in charge

Goes with the AI part of CBSE's Class 7 CT&AI handbook, chapter 4



At a glance	
Students will	explain three kinds of bias, check accuracy for each group and say why humans must review important decisions.
Key words	**bias** (an unfair preference), **fairness** (works equally well for everyone), **transparency** (being open about how it decides), **human in the loop** (a person checks the decision)
Materials	AI picture cards: a lopsided training deck (18 fruit, 2 vegetable), a test deck (10 fruit, 10 vegetable), 8 extra vegetable cards, scoring slips
Time	7 periods of 40 minutes
Link	Goes with the AI part of CBSE's Class 7 CT&AI handbook, chapter 4.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *bias*, *fairness*, *transparency*, *human in the loop*. Work through the example *The lopsided deck* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: The lopsided deck	Run the activity below. Move between groups and note who can explain the idea.	Train a rule on lopsided data, measure accuracy for each group, fix the data and test again.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	A fairness check you can run on any classifier. Model the first card or step, then let pairs finish.	After adding 16 vegetable cards, the classifier gets 46 of 50 fruits and 9 of 10 vegetables right. Work out the accuracy for each group and overall. Is it fairer?
5	Bias detectives	Give case cards (hiring, face recognition, health spending). Groups name the kind of bias: data, historical or measurement.	Name the bias in each case and suggest one fix.
6	Digital citizenship	Discuss a cyberbullying case using a fake account. What should a bystander do?	Agree class rules for respectful online behaviour.
7	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

© 2026 Bullseye Knowledge System Limited

Hands-on activity: The lopsided deck

1. Groups look only at the lopsided training deck and write a rule for 'fruit or vegetable'. They predict their score.
2. Test on the balanced deck. Groups count right answers for fruits and vegetables separately.
3. Give out the 8 extra vegetable cards. Groups rewrite the rule with the more balanced deck and test again.
4. Ask: 'Where could this happen in real life?' (faces, voices, languages)

Tip: Keep the decks apart. The point is that the bias came from the data people chose, not from the 'machine'.

Common mistakes to watch for

- **Trusting the overall score.** A good overall score can hide a group that is mostly wrong.
- **Blaming the machine.** A classifier copies its examples.
- **Fixing by removing** hard cards from the test deck, which hides the problem.

Different paces

Support: Give groups a ready-made rule and ask them only to test it and count.
Stretch: Write accuracy for each group as a percentage, before and after adding vegetable cards, and draw a bar chart of both.

If you have computers (optional)

Train a no-code image model with far more photos of one class than the other, test it, then rebalance and compare.

Exit check and what to note

Two quick questions at the end of the chapter:

1. A classifier gets 45 of 50 fruits and 3 of 10 vegetables right. What is its overall accuracy? (Answer: 48 of 60 = 80%)

2. Why is 80% misleading here? (Answer: vegetables are only 30% right)

Note in the journal: Beginning (needs help to explain three kinds of bias); **Developing** (can explain three kinds of bias with prompts or a partner); **Secure** (can explain three kinds of bias and check accuracy for each group alone); **Extending** (can say why humans must review important decisions and explain why).

© 2026 Bullseye Knowledge System Limited

Answer key

1. b) historical bias
2. a) data bias
3. c) measurement bias: spending is not the same as illness
4. a) a person should review it, because important decisions need a human in the loop
5. a) check it, and never share posts that hurt someone
Build it: Fruit 92%, vegetable 90%, overall 55 of 60 (about 92%). Yes: both groups are now about equally accurate
Challenge: Open answer, for example: a teacher makes the final decision; check that the AI works equally well for every group; explain to students how it was used

© 2026 Bullseye Knowledge System Limited

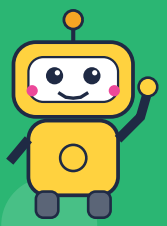
P1

TEACHER GUIDE · PROJECT 1 · 20 HOURS · LEVEL C5

Nutri-Quest Food Planner

Project skill: Use data and rules to plan a balanced week

CBSE: 40 project hours a year in Class 7, as two projects of 20 hours



At a glance	
Big question	Can we use nutrition data and simple rules to plan a balanced, affordable week of school lunches?
Students will	collect nutrition and cost data, write rules for a balanced lunch, plan a week that follows every rule, check and improve it, and present it with charts
Groups	Teams of 3 or 4, with roles that rotate at the halfway point
Materials	Food packets or printed food-composition tables, data sheet pads (kit), squared paper, chart paper, calculators (optional)
Time	30 periods of 40 minutes (about 2 periods a week for 10 weeks)
Evidence	Food data sheet with sources, written rules, two versions of the plan with rule checks, two charts, the presentation and the completed rubric

© 2026 Bullseye Knowledge System Limited

Week by week

Weeks	Stage	Students do	Teacher does
1	Define	Agree the question and the food groups. Decide what data to collect for each food.	Form teams and roles. Agree the food groups and that all values will be per serving.
2	Collect	Record food group, protein, energy and cost per serving for 12 foods. Write each source.	Provide food packets or printed tables (such as ICMR-NIN's Indian Food Composition Tables). Check that every value has a source.
3	Organise	Make one table of all the foods. Check for missing or odd values.	Model one row of the food table. Check units: per serving, not per 100 g.
4	Write rules	Write 3 to 5 rules as 'every lunch must ...' or 'if ... then ...'.	Help teams turn ideas into testable rules ('every lunch must ...'). Agree the rules before planning starts.
5	Build	Make a first plan for Monday to Friday.	Let teams build a first plan quickly; mistakes are useful at this stage.
6	Check	Check each day against each rule in a table. Fix every broken rule.	Model the rule-check table with one day. Check that teams test every rule on every day.
7	Improve	Make version 2: cheaper, more protein or more variety. Compare the two versions.	Ask teams to decide what 'better' means (cheaper, more protein, more variety) and to measure it.
8	Show	Draw a bar chart of protein per day and a pie chart of cost by food group.	Check chart choices: bar charts for protein per day, pie charts for cost by food group.
9	Present	Present the rules, the plan, the checks and the charts in 3 minutes.	Run 3-minute presentations with 'two stars and a wish' feedback.
10	Reflect	Score your work with the rubric and write what you would change.	Students score themselves with the rubric; you score too. Discuss any differences.

Setting up

- ✓ Collect food data only from packets or trusted tables, with a source for every value.
- ✓ Do not collect anyone's weight, height or health details.
- ✓ Respect religious, cultural and health food choices, including allergies.
- ✓ Agree the budget and the rules before planning starts, and do not change them to make a plan pass.

© 2026 Bullseye Knowledge System Limited

★ Team roles

Give each student a role and rotate roles at the halfway point, so everyone collects, checks, builds and presents.

Data manager: keeps the food table and sources.

Rule checker: checks every day against every rule.

Planner: builds and improves the weekly plan.

Presenter: prepares the charts and leads the talk.

Common pitfalls

- ✓ **Mixing units**: per 100 g on a packet and per serving in the plan.
- ✓ **Checking only some rules**, or only some days.
- ✓ **Changing a rule** to make a plan pass instead of changing the plan.

Different paces

Support: Give a ready-made food table and three rules; teams build and check the plan.
Stretch: Add a rule for variety (at least 4 different vegetables in the week) and find the cheapest plan that passes every rule.

If you have computers (optional)

Use a spreadsheet with formulas that flag a broken rule automatically, such as a cell that turns red when a lunch costs more than Rs 40.

Marking with the rubric

Score each row from 1 to 4 using the data sheet, rule checks, both plan versions, charts and talk; the total is out of 24. Give credit for honest checks that found broken rules.

What we check	What excellent (4) looks like
Question and rules	Clear rules that make a lunch balanced and affordable.
Data	Nutrition and cost data for every food, each with a source.
Plan	Every day follows every rule, and the checks are shown.
Improvement	Version 2 is better in a way that can be measured.
Charts and talk	The right chart for each job; a clear, short talk.
Teamwork	Everyone has a job and the work is shared fairly.

© 2026 Bullseye Knowledge System Limited

What a strong project sounds like

'Version 1 broke two rules: dal came three times and Wednesday cost Rs 46. Version 2 swaps in chana, costs Rs 34 a day on average, and passes all four rules every day. We checked each day in a table. Our plan is a school project, not medical advice.'

© 2026 Bullseye Knowledge System Limited

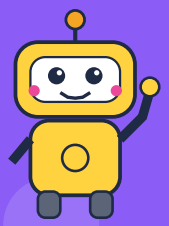
P2

TEACHER GUIDE · PROJECT 2 · 20 HOURS · LEVEL C5

Sort It with AI

Project skill: Build a waste classifier and measure its accuracy

CBSE: 40 project hours a year in Class 7, as two projects of 20 hours



At a glance	
Big question	Can we teach a classifier to put school waste in the right bin, and measure how often it is right?
Students will	make labelled picture cards of waste, build a decision tree that sorts them, test it on unseen cards, measure accuracy, and improve it
Groups	Teams of 3 or 4, with roles that rotate at the halfway point
Materials	Blank cards and pencils (or printed waste pictures), the AI picture cards (kit), data sheet pads, chart paper
Time	30 periods of 40 minutes (about 2 periods a week for 10 weeks)
Evidence	60 labelled cards, the decision tree, the test record, accuracy before and after improving, a chart and the completed rubric

© 2026 Bullseye Knowledge System Limited

Week by week

Weeks	Stage	Students do	Teacher does
1	Define	Agree the question and the three bins. Decide how you will measure success.	Form teams and roles. Agree the three bins (wet, dry, hazardous) and that success means accuracy on unseen cards.
2	Collect	Make 60 picture cards of waste items, about 20 for each bin, and label each one.	Teams make 60 picture cards, about 20 per bin. Use pictures only: no real waste in class.
3	Check labels	Check unclear labels with your teacher. Shuffle and split: 40 training, 20 test.	Check unclear labels with each team. Shuffle and split 40 training and 20 test cards; collect the test cards and keep them.
4	Find features	Study the training cards: what can you observe that separates the bins?	Ask teams to list features they can observe, such as 'made from food or plants' or 'contains a battery'.
5	Build	Write your rules as a decision tree with yes/no questions.	Check that every path in the tree ends in a bin and that each question is yes/no.
6	Test	Run the tree on the 20 test cards. Record each prediction and the true label.	Hand back the test cards. Teams record each prediction and the true bin.
7	Measure	Work out the accuracy. Make a table of which bins were mixed up.	Check accuracy calculations and ask for accuracy for each bin, not only overall.
8	Improve	Fix a rule or add training cards, then test on new cards you have not used.	Teams fix a rule or add training cards, then test on new cards you provide.
9	Present	Show the tree, the results before and after, and what you learnt.	Run presentations: the tree, results before and after, and lessons learnt.
10	Reflect	Score your work with the rubric. Which mistakes matter most?	Students score themselves with the rubric; you score too. Discuss which mistakes matter most.

Setting up

- ✓ Use pictures only. Never handle real broken glass, batteries, medicines or dirty waste in class.
- ✓ Keep the test cards with you until the testing week.
- ✓ Agree the three bins and their meanings using your local waste rules.
- ✓ Prepare 10 to 15 extra 'new' cards for the improvement test.

© 2026 Bullseye Knowledge System Limited

★ Team roles

Give each student a role and rotate roles at the halfway point, so everyone collects, checks, builds and presents.

Card maker: draws and labels the cards and checks labels.

Tree designer: writes the decision-tree questions.

Tester: runs the tree on test cards and records results.

Analyst: works out accuracy and leads the presentation.

Common pitfalls

- ✓ **Test cards leaking into training**, which makes accuracy look too good.
- ✓ **Accuracy worked out on some cards only**, or as a count instead of a fraction or percentage.
- ✓ **Reporting only overall accuracy** and hiding a weak bin.

Different paces

Support: Give a two-question starter tree and 30 ready-made cards.
Stretch: Add a fourth bin (e-waste) and show how the tree and accuracy change.

If you have computers (optional)

Train a no-code image model on photos of the same waste items, test it on new photos and compare its accuracy with the team's decision tree.

Marking with the rubric

Score each row from 1 to 4 using the cards, tree, test record and talk; the total is out of 24. Check that accuracy is worked out on unseen cards; give credit for honest reporting of weak bins.

What we check	What excellent (4) looks like
Data	60 clearly labelled cards; labels checked; test cards kept apart.
Rules	A clear decision tree that uses features anyone can observe.
Testing	Accuracy worked out correctly on unseen cards; mistakes listed.
Improvement	A rule or the data is improved, and accuracy is measured again.
Fairness and honesty	Real results reported, and accuracy checked for each bin.
Talk and teamwork	A clear presentation; the work is shared fairly.

© 2026 Bullseye Knowledge System Limited

What a strong project sounds like

'Our tree got 17 of 20 test cards right, which is 85%. It was weakest on hazardous waste, 4 of 6. We added a question about batteries and thermometers and tested again on 12 new cards: 11 right. Putting a battery in the wet bin is the most harmful mistake.'

© 2026 Bullseye Knowledge System Limited

Observation journal

Class 7 ____ Section ____ Term ____ Chapters _____. Circle one level per skill: **B**eginning, **D**eveloping, **S**ecure, **E**xtending. You may photocopy this page.

Student	Algorithms	Logic	AI ideas	Projects	Notes
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	

© 2026 Bullseye Knowledge System Limited

Term summary for the report card

One copy per child each term. Fill it from the observation journal and the child's self-checks. You may photocopy this page.

Name _____ Class _____ Section _____ Term _____

Area	Level (circle one)	Teacher comment
Computational thinking (overall)	B D S E	
Algorithms, formulas and binary	B D S E	
Logic and checking	B D S E	
AI ideas, data and fairness	B D S E	
Projects (rubric score)	B D S E	

Writing comments

Write one strength and one next step, in words a parent understands. For example: *Explains steps clearly and fixes bugs in routes; next step: spotting the core of a pattern.*

Class teacher _____ Date _____

© 2026 Bullseye Knowledge System Limited